Materials Needed from the University
Professional Structural Vibration Analyzer

Purpose: Equipment request for validating a DIY ESP32/MPU6050 vibration analyzer against a professional measurement system.

Objective

The objective of this equipment request is to assemble a professional structural vibration measurement system that can be used as a reference to compare against a low-cost DIY vibration analyzer based on an ESP32 and MPU6050 accelerometers. The professional system will be used to measure vibration signals with higher accuracy, better resolution, proper sensor calibration, and reliable frequency-domain analysis.

The requested equipment will allow comparison of:

- Time-domain acceleration response
- RMS acceleration
- Peak acceleration
- Dominant frequency from FFT analysis
- Frequency response behavior
- Noise floor and sensor sensitivity
- Repeatability of measurements
- Limitations of low-cost accelerometer-based systems

Required Materials

Material / Equipment	**Description and Reason for Use**
NI CompactDAQ Chassis	A National Instruments CompactDAQ chassis, such as a cDAQ-9171, cDAQ-9174, cDAQ-9178, or similar, is needed to connect the vibration input module to the computer through USB or another supported interface. This chassis acts as the main data acquisition platform and allows LabVIEW to communicate with the measurement hardware.
NI-9234 Dynamic Signal Acquisition Module	The NI-9234 is the preferred vibration input module because it is designed for dynamic signal measurements using accelerometers. It provides simultaneous sampling, high resolution, built-in anti-aliasing filters, and IEPE/ICP sensor excitation. This is necessary to obtain professional-quality vibration data and compare it against the DIY MPU6050-based system.
NI-9231 or Equivalent IEPE Module	If the NI-9234 is not available, an NI-9231 or another equivalent IEPE-compatible dynamic signal acquisition module may be used. The important requirement is that the module supports accelerometer measurements, high sampling rates, and IEPE/ICP excitation. This would allow the system to still function as a professional reference analyzer.
IEPE/ICP Accelerometer	One or more professional IEPE/ICP accelerometers are needed to measure vibration accurately. These sensors are commonly used in industry and research for structural vibration, machinery diagnostics, and modal testing. They are necessary because they provide better sensitivity, lower noise, and wider bandwidth than low-cost MEMS accelerometers such as the MPU6050.
Triaxial IEPE Accelerometer	A triaxial accelerometer is preferred if available because it can measure acceleration in the X, Y, and Z directions at the same mounting location. This would allow a direct comparison with the MPU6050, which is also a three-axis accelerometer. It would reduce the need to repeat the test multiple times for different sensor orientations.
Single-Axis IEPE Accelerometer	If a triaxial accelerometer is not available, a single-axis IEPE accelerometer can still be used. The sensor can be mounted in one direction at a time to compare each axis of the DIY system separately. This is sufficient for validating the dominant frequencies, amplitude behavior, and general accuracy of the DIY vibration analyzer.

Material / Equipment	Description and Reason for Use
Accelerometer Cables	Compatible accelerometer cables are required to connect the IEPE/ICP accelerometers to the NI input module. The cable type depends on the accelerometer connector and may include BNC-to-10-32, BNC-to-microdot, or triaxial accelerometer cables. These cables are necessary because regular wires are not suitable for reliable low-noise vibration measurements.
BNC Cables	BNC cables may be needed to connect sensors, signal conditioners, or test equipment to the NI input module. They provide a secure and shielded connection, which helps reduce electrical noise in the vibration signal.
Accelerometer Mounting Studs	Mounting studs are used to rigidly attach the accelerometer to the structure being tested. Stud mounting is one of the best methods for vibration testing because it provides a stiff connection and improves high-frequency measurement accuracy.
Adhesive Mounting Pads	Adhesive pads may be used when drilling or threaded mounting is not possible. They are useful for temporary mounting on the 3D printer frame, printer head, or other test structures. The mounting must be as rigid as possible to avoid distorting the measured vibration response.
Mounting Wax	Mounting wax is useful for temporary accelerometer installation during experimental testing. It allows quick sensor placement and removal while maintaining a reasonably rigid connection. This is helpful when testing multiple locations on the printer frame or other structures.
Magnetic Accelerometer Base	A magnetic base can be used to mount accelerometers to steel or ferromagnetic surfaces without drilling. It is useful for quick testing, but it should only be used when the surface is compatible and when the frequency range of interest is not affected by the magnetic mount.
Small Mounting Block or Adapter	A mounting block or adapter may be required to place the accelerometer close to the DIY MPU6050 sensor. This helps align both sensors at the same location and orientation, which is important for a fair comparison between the professional and DIY systems.

Material / Equipment	Description and Reason for Use
Accelerometer Calibrator	An accelerometer calibrator or handheld vibration shaker is highly recommended. It produces a known vibration level, often at a fixed frequency and acceleration amplitude. This is necessary to verify that the professional accelerometer and acquisition system are reading correctly before they are used as the reference measurement system.
Computer with LabVIEW Installed	A computer with LabVIEW is needed to acquire, display, and process the vibration data from the NI hardware. Since the project is intended to use LabVIEW as the main professional software environment, access to a properly configured computer is required.
NI-DAQmx Driver	NI-DAQmx is required for LabVIEW to communicate with the NI CompactDAQ hardware. It allows the user to configure channels, sampling rate, sensor sensitivity, input range, and data acquisition settings. Without NI-DAQmx, the NI hardware cannot be properly controlled from LabVIEW.
NI Measurement & Automation Explorer	NI MAX is useful for checking whether the NI chassis and input modules are detected correctly. It can also be used to test channels, verify sensor input, configure devices, and troubleshoot hardware communication before running the LabVIEW program.
NI Sound and Vibration Toolkit	The Sound and Vibration Toolkit is preferred because it includes tools for vibration analysis, frequency analysis, power spectrum calculation, frequency response functions, octave analysis, and other dynamic signal processing tasks. It is not strictly required, but it would make the professional analyzer easier to build and more complete.
MATLAB or Python Access	MATLAB or Python can be used for additional post-processing of the vibration data. This is useful because the DIY analyzer already uses Python-based FFT analysis, so the same or similar analysis methods can be applied to both systems for a fair comparison.
Modal Hammer	A modal or instrumented impact hammer is optional, but useful if impact testing or modal analysis will be performed. It can provide a controlled input force to the structure, allowing frequency response functions and natural frequencies to be identified more clearly.

Material / Equipment	Description and Reason for Use
Force Sensor or Load Cell	A force sensor is optional and would only be needed if the test requires measuring the input force applied to the structure. This would be useful for advanced modal testing where both force input and acceleration output are measured.
Test Structure or Rigid Mounting Surface	A stable test structure is needed to mount the sensors and perform controlled vibration testing. This may be the 3D printer frame, a beam, a plate, or another mechanical structure. The structure should allow the professional accelerometer and DIY sensor to be mounted close together.
Function Generator	A function generator is optional but useful if controlled sinusoidal excitation is needed. It can generate known frequencies that can be sent to a shaker, speaker, or other actuator. This helps test whether both the professional and DIY systems can detect the same forced vibration frequency.
Shaker or Excitation Source	A shaker, speaker-based exciter, or another vibration source may be used to excite the structure at controlled frequencies. This is necessary if the goal is to test the analyzer using repeatable vibration inputs instead of relying only on the natural operation of the 3D printer or machine.
Power Supply	A power supply may be needed for external excitation devices, signal conditioning equipment, or any additional electronics used during the experiment. Stable power is important to avoid introducing noise or inconsistent vibration behavior.
USB Cables and Power Cables	USB and power cables are needed to connect the CompactDAQ chassis, computer, and any additional equipment. These should be included to avoid setup delays during the experiment.
Measuring Tools	Calipers, ruler, tape measure, and basic hand tools are useful for documenting sensor locations, mounting distances, and structure dimensions. This information is important for writing a complete experimental methodology.
Camera or Phone for Documentation	A camera or phone should be used to document the sensor placement, wiring, mounting method, and test setup. Photographs are important for reports and help show that the professional and DIY sensors were tested under the same conditions.

Minimum Required Setup

If the university cannot provide all the listed equipment, the minimum useful professional setup would be:

- NI CompactDAQ chassis
- NI-9234 or equivalent IEPE dynamic signal acquisition module
- One IEPE/ICP accelerometer
- Compatible accelerometer cable
- Basic mounting hardware
- Computer with LabVIEW and NI-DAQmx

This minimum setup would allow one professional accelerometer to be mounted beside one DIY MPU6050 sensor. The same vibration event could then be recorded by both systems and compared in the time domain and frequency domain.

Preferred Setup

The preferred setup for a more complete validation would be:

- NI CompactDAQ chassis
- NI-9234 dynamic signal acquisition module
- One triaxial IEPE accelerometer or three single-axis IEPE accelerometers
- Compatible accelerometer cables
- Accelerometer mounting kit
- Accelerometer calibrator
- Computer with LabVIEW, NI-DAQmx, NI MAX, and Sound and Vibration Toolkit
- MATLAB or Python for additional post-processing

This preferred setup would allow comparison of three-axis vibration data from the professional system with the three-axis MPU6050 data from the DIY analyzer.

Optional Equipment for Advanced Testing

The following equipment is not required for the first validation test, but would be useful if the project expands into modal analysis or more controlled structural testing:

- Instrumented modal hammer
- Force sensor or load cell
- Electrodynamic shaker
- Function generator
- Power amplifier
- Additional accelerometers
- Laser tachometer for rotating machinery tests

Recommended Loan Request Statement

The following statement can be used when requesting the equipment from the university:

> I would like to request access to a professional vibration measurement setup to validate a low-cost DIY structural vibration analyzer based on an ESP32 and MPU6050 accelerometers. Ideally, I would need an NI CompactDAQ chassis, an NI-9234 or equivalent IEPE dynamic signal acquisition module, one or more IEPE/ICP accelerometers, compatible accelerometer cables, mounting hardware, and access to a computer with LabVIEW and NI-DAQmx. If available, an accelerometer calibrator and the NI Sound and Vibration Toolkit would also be very useful for improving the accuracy and credibility of the validation.

Expected Use of the Equipment

The professional vibration analyzer will be used as a reference system. The IEPE accelerometer will be mounted as close as possible to the DIY MPU6050 sensor on the same structure. Both systems will record the same vibration event, and the results will be compared using:

- Time-domain acceleration plots
- FFT plots
- Dominant frequency identification
- RMS acceleration values

- Peak acceleration values
- Noise level comparison
- Repeatability across multiple trials

This comparison will help determine the accuracy, limitations, and educational usefulness of a low-cost accelerometer-based vibration analyzer.